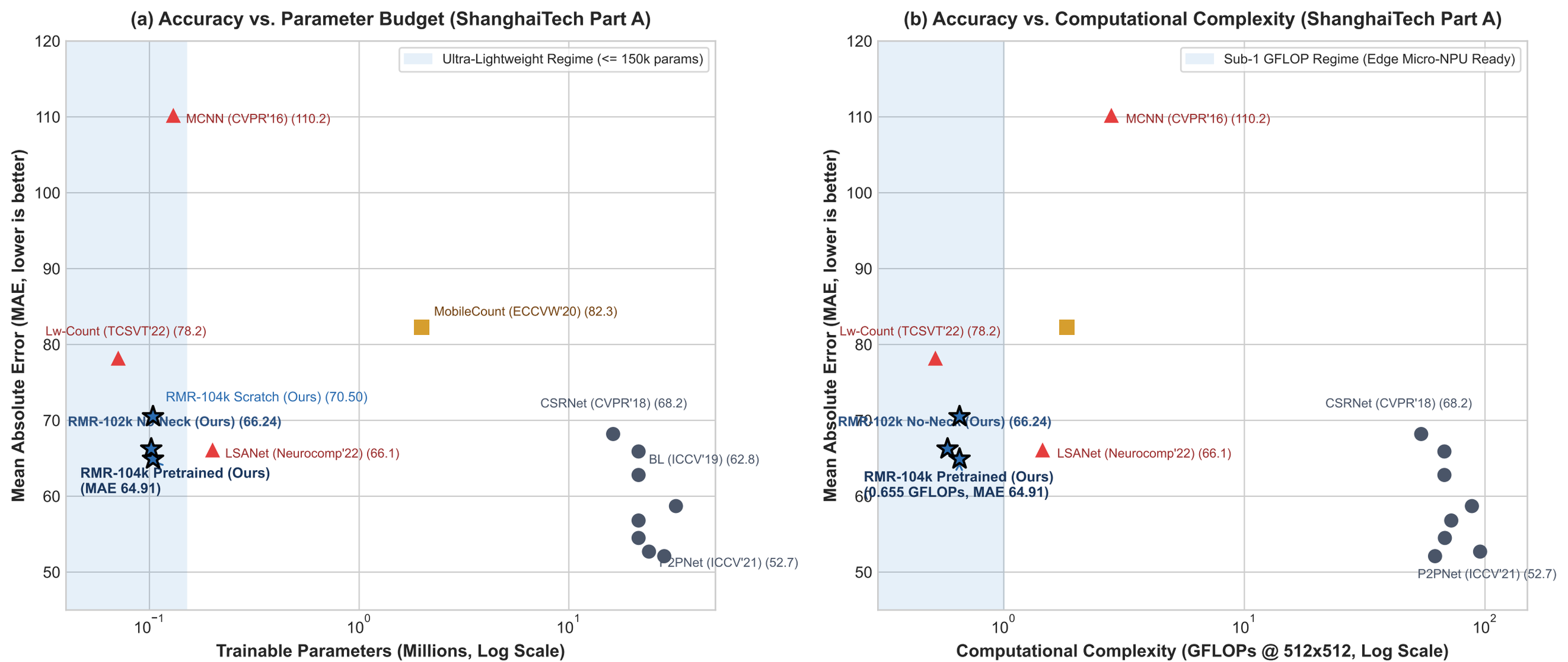
(a) Accuracy vs. Parameter Budget (ShanghaiTech Part A)
Ultra-Lightweight Regime (<= 150k params)
MCNN (CVPR'16) (110.2)
MobileCount (ECCVW'20) (82.3)
Lw-Count (TCSVT'22) (78.2)
RMR-104k Scratch (Ours) (70.50)
RMR-102k No-Neck (Ours) (66.24)
LSANet (Neurocomp'22) (66.1)
RMR-104k Pretrained (Ours)
(MAE 64.91)
CSRNet (CVPR'18) (68.2)
BL (ICCV'19) (62.8)
P2PNet (ICCV'21) (52.7)
Mean Absolute Error (MAE, lower is better)
Trainable Parameters (Millions, Log Scale)
(b) Accuracy vs. Computational Complexity (ShanghaiTech Part A)
Sub-1 GFLOP Regime (Edge Micro-NPU Ready)
MCNN (CVPR'16) (110.2)
Lw-Count (TCSVT'22) (78.2)
RMR-102k No-Neck (Ours) (66.24)
LSANet (Neurocomp'22) (66.1)
RMR-104k Pretrained (Ours)
(0.655 GFLOPs, MAE 64.91)
CSRNet (CVPR'18) (68.2)
P2PNet (ICCV'21) (52.7)
Mean Absolute Error (MAE, lower is better)
Computational Complexity (GFLOPs @ 512x512, Log Scale)